large shift indicates the component is critical for the prediction, while a small shift suggests a limited role. We quantify each shift using Jensen-Shannon (JS) divergence between the original and intervened output distributions. Specifically, for subject-token and last-token representations, we corrupt the corresponding hidden states at each layer ℓ. For the attention mechanism, we follow Geva et al. (2023) to mask the attention flow between the subject and last tokens at layer ℓ, using a window size of 5 layers.

Experimental Results As shown in Figure 2, FAs and AHs share highly similar information flows. The components that are crucial for producing factually correct outputs are also crucial for producing AHs. This pattern suggests that, despite being factually incorrect, AHs are generated through the same underlying internal mechanisms as factual recall. In contrast, UHs display substantially weaker information flow, reflecting their limited reliance on the subject entity during generation.

These findings indicate that hidden states primarily capture **whether the model is recalling parametric knowledge**, rather than **whether the generated output is factually correct**. These results challenge the prevailing assumption in prior works (Azaria and Mitchell, 2023; Gottesman and Geva, 2024; Yüksekgönül et al., 2024; Orgad et al., 2025; Varshney et al., 2023b) that internal representations inherently encode factual correctness. Instead, our results show that factual outputs and association-driven hallucinations can share nearly identical internal computational pathways.

It is worth noting that the strong causal effect of attention-flow interventions for FAs and AHs is expected, since our labeling procedure already uses attention masking to distinguish AHs from UHs. However, interventions on subject-token representations and last-token hidden states provide an independent validation of our hypothesis. These components remain highly influential for both FAs and AHs, confirming that the shared mechanism extends beyond the attention pathways used during labeling.

4.2 Analysis of Subject Representations

Having established the overall information flow, we next analyze the subject representations in the early layers. Our goal is to understand why the internal behavior at subject positions is highly similar for AHs and FAs, yet differs substantially for UHs. To this end, we first examine the norm of these representations (§4.2.1), and then analyze how this behavior correlates with subject prevalence in the pre-training data (§4.2.2).

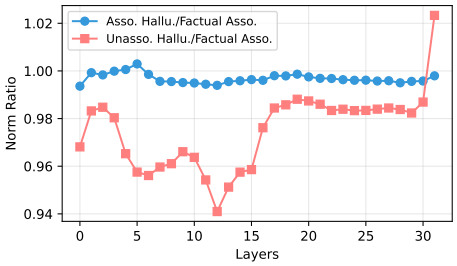


Figure 3: Norm ratio curves of subject representations in LLaMA-3-8B, comparing AHs and UHs against FAs as the baseline.

4.2.1 Norm of Subject Representations

We measure the average L_2 norm of subject representations for each knowledge category across layers (details in Appendix B). To facilitate direct comparison, we report the L_2 norms of both hallucination types (AHs and UHs) as ratios relative to the norm of factually correct predictions (FAs). A ratio close to 1 indicates norms comparable to FAs, while ratios below 1 indicate smaller norms.

As shown in Figure 3, for LLaMA-3-8B, the norm of AH samples is highly comparable to that of FAs, with ratios remaining close to 1 across layers. In contrast, UH samples consistently exhibit smaller norms than FAs in nearly all layers, with the ratio dropping to approximately 0.95 in the middle layers (layers 10–15). This suggests that **AHs and FAs activate subject representations with similar strength, whereas UHs involve weaker subject encoding**. We further analyze the subspace overlap between subject hidden states and MLP weight matrices in Appendix D.1, which provides additional evidence that AHs and FAs occupy highly similar representational regions.

4.2.2 Correlation with Subject Popularity

To explain why AHs exhibit subject representation norms comparable to FAs while UHs remain weaker, we hypothesize that this difference arises from subject popularity in the pre-training data. Prior work (Kang et al., 2024) shows that entities appearing more frequently during pre-training tend to produce hidden representations with larger norms. Accordingly, we expect that FAs and AHs predominantly involve high-popularity entities,

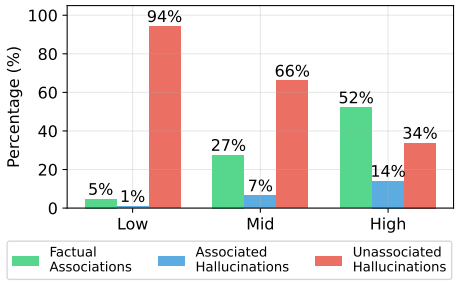


Figure 4: Sample distribution across different subject popularity (low, mid, high) in LLaMA-3-8B, measured by monthly Wikipedia page views.

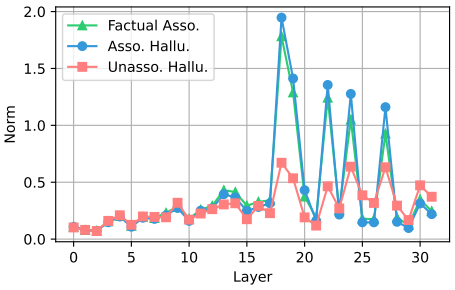


Figure 5: Subject-to-last attention contribution norms across layers in LLaMA-3-8B. Values show the norm of the attention contribution from subject tokens to the last token at each layer.

while UHs are more common for low-popularity entities.

Following Mallen et al. (2023a), we use average monthly Wikipedia page views as a proxy for subject frequency during pre-training. We partition subjects into three popularity bins and analyze the distribution of FAs, AHs, and UHs across them. The distribution in Figure 4 confirms that the prevalence of both FAs and AHs is strongly associated with subject popularity. In the lowest popularity bin, both FAs and AHs are rarely observed (5% for FAs and 1% for AHs). In contrast, their prevalence significantly increases in the highest popularity bin (52% for FAs and 14% for AHs). However, the prevalence of UHs shows an inverse trend: while the lowest popularity bin is dominated by UHs (94%), their prevalence drops to 34% in the highest popularity bin.

This observation highlights a critical implication: if high-popularity subjects account for a substantial portion of real-world language model usage, AHs may constitute a substantial portion of hallucinations encountered in practical deployments. Meanwhile, as AHs arise mainly on popular subjects, they are often indistinguishable from FAs by popularity-based heuristics, contradicting prior work (Mallen et al., 2023a) that links popularity to hallucinations.

4.3 Visualization of Attention Flow

Having examined how the model forms subject representations, we next visualize how this information is propagated to the last token of the input where the model generates the object of a knowledge tuple. While the strong reliance of AHs on this attention flow is an expected consequence of our causal intervention labeling, visualizing this mechanism explicitly illustrates the divergence between AHs/FAs and UHs, providing a more comprehensive view of the internal mechanism behind how the model produces outputs for each category.

To quantify the specific contribution from subject tokens $(s_1, ..., s_n)$ to the last token, we compute the attention contribution from subject tokens to the last position:

$$\mathbf{a}_{\text{last}}^{\ell} = \sum_k \sum_h A_{\text{last},s_k}^{\ell,h} (\mathbf{h}_{s_k}^{\ell-1} W_V^{\ell,h}) W_O^{\ell,h}. \quad (1)$$

where $A_{i,j}^{\ell,h}$ denotes the attention weight assigned by the h-th head in the layer ℓ from the last position i to subject token j. Here, $\mathbf{a}_{\text{last}}^{\ell}$ represents the subject-to-last attention contribution at layer ℓ. Intuitively, if subject information is critical for prediction, this contribution should have a large norm; otherwise, the norm should be small.

Figure 5 shows that in LLaMA-3-8B, both AHs and FAs exhibit large attention-contribution norms in mid-layers, indicating a strong information flow from subject tokens to the target token. In contrast, UHs show consistently lower norms, implying that their predictions rely far less on subject information. Yüksekgönül et al. (2024) previously argued that high attention flow from subject tokens signals factuality and proposed using attention-based hidden states to detect hallucinations. Our results challenge this view: **the model propagates subject information just as strongly when generating AHs as when producing correct facts**.

4.4 Analysis of Last Token Representations

Since both FAs and AHs involve strong information propagation from subject representations, whereas this propagation is weaker for UHs, we hypothesize that this disparity in information flow also leads to distinct geometric properties in the last token representations. Specifically, the strong subject propagation for FAs and AHs suggests that their last-token representations receive significant subject-specific updates, causing them to ***diverge*** in

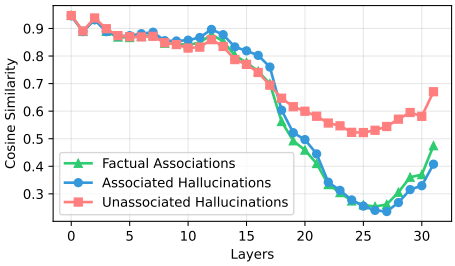


Figure 6: Cosine similarity of target-token hidden states across layers in LLaMA-3-8B.

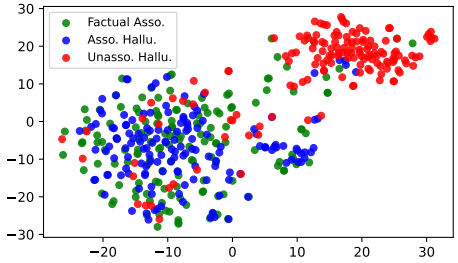


Figure 7: t-SNE visualization of last token's representations at layer 25 of LLaMA-3-8B.

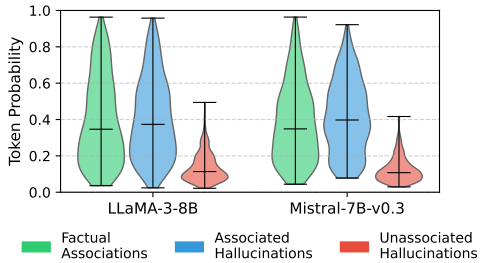


Figure 8: Distribution of last token probabilities.

representation space. In contrast, because subject information is weakly propagated for UH samples, their last-token representations receive smaller updates and therefore are expected to remain ***tightly clustered***, dominated by the shared features of the prompt template.

To verify this, we compute the layer-wise cosine similarity among last-token representations $\mathbf{h}_T^\ell$ within each knowledge category (Figure 6). The results indicate that the strength of propagated subject information significantly affects the geometric properties. In early layers, cosine similarity is high for all categories ($\approx$ 0.9). From the mid-layers onward, where subject representations propagate for FAs and AHs, their last-token representations diverge significantly. By layer 25, the cosine similarity for both FAs and AHs drops to $\approx$ 0.2. In contrast, UHs remain moderately clustered, with cosine similarity only declining to $\approx$ 0.5.

To further illustrate this, we present a t-SNE visualization of the last token's representations at layer 25 of LLaMA-3-8B in Figure 7. The visualization aligns with our cosine similarity analysis: hidden representations of UHs form a distinct cluster separated from FAs, whereas AHs substantially overlap with FAs. We quantitatively validate this observation using two cluster separability metrics (the Silhouette coefficient and Davies-Bouldin Index) in Appendix D.2, which show significantly higher separability for UHs compared to AHs. Additional visualizations can be found in Appendix D.3.

Additionally, we observe that the strength of subject representation propagation also affects the entropy of the output distribution (Figure 8). A strong subject-specific update concentrates the output distribution on specific subject-related entities, resulting in low-entropy outputs for FAs and AHs. Conversely, a weak subject-specific update leads to a wider spread of probability mass covering many possible answers (e.g., assigning probability to multiple possible names for the prompt "*The name of the father of <subject> is*"), resulting in significantly higher output entropy.

5 Revisiting Hallucination Detection

The mechanistic analysis in §4 reveals that *internal states of LLMs primarily capture* ***how the model utilizes its parametric knowledge****, rather than whether the output is truthful*. This suggests that representation-based signals struggle to distinguish associated hallucinations (AHs) from factual associations (FAs) but can more effectively separate unassociated hallucinations (UHs) from FAs.

In this section, we quantify the extent to which existing hallucination detection methods can separate AHs from FAs and compare this performance to detecting UHs.

Experimental Setups We revisit the effectiveness of widely-adopted white-box hallucination detection approaches that use internal state probing as well as black-box approaches that rely on scalar features. We evaluate on three settings: 1) **AH Only** (1,000 FAs and 1,000 AHs for training; 200 of each for testing), 2) **UH Only** (1,000 FAs and 1,000 UHs for training; 200 of each for testing), and 3) **Full** (1,000 FAs and 1,000 hallucination samples mixed of AHs and UHs for training; 200 of each for testing). For each setting, we repeat data splitting with five random seeds and report mean AUROC and standard deviation.

White-box methods: We extract and normalize

Methods	LLaMA		Mistral	
	AH Only	UH Only	AH Only	UH Only
Subject	0.65 ± 0.02	0.91 ± 0.01	0.57 ± 0.02	0.81 ± 0.02
Attention	0.58 ± 0.04	0.92 ± 0.02	0.58 ± 0.07	0.87 ± 0.01
Last Token	$\mathbf{0.69 \pm 0.03}$	$\mathbf{0.93 \pm 0.01}$	$\mathbf{0.63 \pm 0.02}$	$\mathbf{0.92 \pm 0.01}$
Probability	0.49 ± 0.01	0.86 ± 0.01	0.46 ± 0.00	0.89 ± 0.00
Subject Pop.	0.48 ± 0.01	0.87 ± 0.01	0.52 ± 0.01	0.84 ± 0.01

Table 2: Hallucination detection performance on **AH Only** and **UH Only** settings. Detailed statistical significance analysis is provided in Appendix E.1.

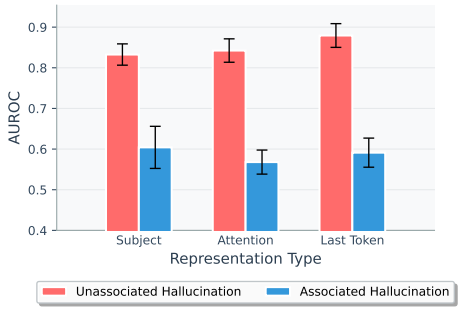


Figure 9: Hallucination detection performance on the **Full** setting (LLaMA-3-8B).

internal features and then train a probe.

- **Subject representations**: last subject token hidden state from three consecutive layers (Gottesman and Geva, 2024).
- **Attention flow**: attention weights from the last token to subject tokens across all layers (Yüksekgönül et al., 2024).
- **Last-token representations**: final token hidden state from the last layer (Orgad et al., 2025).

Black-box methods: We test two commonly used scalar features, including **answer token probability** (Orgad et al., 2025) and **subject popularity** (average monthly Wikipedia page views) (Mallen et al., 2023a). As discussed in §4.2.2 and §4.4, these features are also related to whether the model relies on encoded knowledge to produce outputs rather than with truthfulness itself.

Experimental Results Table 2 shows that hallucination detection methods perform differently in **AH Only** and **UH Only** settings. For white-box probes, all approaches effectively distinguish UHs from FAs (AUROC $\approx$ 0.91–0.93). In contrast, performance drops sharply on the **AH Only** setting (AUROC $\approx$ 0.58–0.69). Black-box methods follow the same pattern (AUROC $\approx$ 0.86–0.87 on **UH Only** and $\approx$ 0.48–0.49 on **AH Only**). Notably, the performance of black-box methods on **AH Only** is close to random guessing (AUROC $\approx$ 0.48–0.49), indicating that token uncertainty and subject popularity provide little signal for separating AHs from

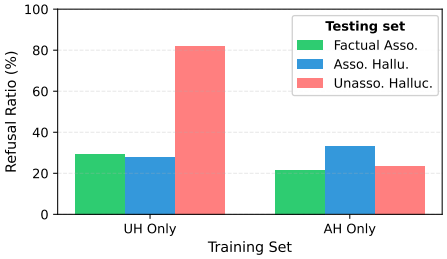


Figure 10: Refusal tuning performance across three types of samples (LLaMA-3-8B).

FAs. Although white-box representations show slightly better separability than black-box methods, such performance is far from reliable for deployment (Hosmer Jr et al., 2013). Figure 9 further highlights this disparity under the **Full** setting: detection is consistently stronger on UH samples than on AH samples, and adding AHs to the training set significantly dilutes performance on UHs (AUROC $\approx$ 0.9 on **UH Only** vs. $\approx$ 0.8 on **Full**).

These results confirm that both internal probes and black-box methods capture whether a model draws on parametric knowledge, not whether its outputs are factually correct. Our further analysis in Appendix E.2 reveals that consistency-based detection methods also struggle to distinguish AHs from FAs. UHs are easier to detect because they bypass this knowledge, while associated hallucinations are produced through the same recall process as FAs, leaving limited internal cues to distinguish them. An analysis of sensitivity-specificity trade-offs in Appendix E.3 further confirms this limitation, showing that while UHs can be detected with high sensitivity and specificity simultaneously, AH detection requires a significant trade-off between them. As a result, **LLMs appear to have limited intrinsic awareness of their own truthfulness, and detection methods relying on these signals risk misclassifying AHs as correct**, fostering harmful overconfidence in model outputs.

6 Challenges of Refusal Tuning

A common strategy to mitigate potential hallucinations in the model's responses is to fine-tune LLMs to refuse answering when they cannot provide a factual response, *e.g.*, **Refusal Tuning** (Zhang et al., 2024). For such refusal capability to generalize, the training data must contain a shared feature pattern across hallucinated outputs, allowing the model to learn and apply it to unseen cases.

Our analysis in the previous sections shows that